Code No: 114CN **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, February -2024
COMPUTER ORGANIZATION
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A

(25 Marks)

1.a) Define instruction formats. [2]
b) How do addressing modes influence the execution of instructions? [3]
c) What is meant by asynchronous data transfer? [2]
d) What triggers interrupt initiated I/O? [3]
e) Define virtual memory. [2]
f) How is memory connected to the CPU in a computer system? [3]
g) Which addressing mode is commonly used for accessing data stored in consecutive memory locations? [2]
h) How does the 8086 handle memory segmentation? [3]
i) Which register is often used as an index register in 8086 assembly language programs? [2]
j) What is the advantage of using the LOOP instruction in assembly language programs? [3]

PART - B

(50 Marks)

2.a) Give an in-depth overview of memory and its functions in a computer.
b) Elaborate on the significance of status bit conditions in program control. [5+5]

OR

3. What are program interrupts? Differentiate between the types of interrupts in computer organization, giving examples of each. [10]

4. Discuss the modes of transfer in asynchronous serial transfer. Highlight the advantages and disadvantages of each mode. [10]

OR

5.a) Discuss the features and applications of the Intel 8089 IOP.
b) Differentiate between isolated I/O and memory-mapped I/O. [6+4]

6.a) Provide a detailed explanation of the Memory Hierarchy and its significance in computer systems.
b) Differentiate between associative and set associative mapping. [6+4]

OR

7.a) Describe the concept of miss and hit ratio in cache memory. How do these ratios impact system performance?
b) State and explain the factors influencing the access time. [6+4]

8. Explain the special functions of general purpose registers in the 8086 microprocessor. [10]

OR

9. Explain the following terms.
   a) 8086 CPU pin diagram.
   b) Addressing modes of 8086. [5+5]

10. Describe the instruction format of the 8086 microprocessor with examples. [10]

OR

11. Write an assembly program in 8086 to evaluate a simple arithmetic expression involving multiplication, and division. [10]

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